Name: _____ Group: _____

CE Lab: Plates, Peak Shape and the Sample's Own Conductivity

Team roles for today (rotate from last session): Manager keeps the group on the current question, with no running ahead to the simulator. Recorder writes the group's predictions and reconciliations on this sheet. Presenter reports to the class at the checkpoints. Reflector notes where the group's thinking changed.

How this is graded: full credit requires three things for every numbered question: your **written prediction before touching any control**, the **observed result**, and, where they disagree, a sentence on **what changed in your thinking**. A correct answer with no prediction earns partial credit. A wrong prediction honestly reconciled earns full credit.

Part	Topic	Minutes
Getting started	Conditions, current and conductivity	5
A	Where do the plates come from?	13
B	Peak shape from the sample's own conductivity	11
C	A salty sample	8
D	pK_a, pH and migration order	6
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Getting started (5 min)

Open drylab and choose the **CE** tab. Change only these settings:

- **Grid points:** 257. Sharp zone edges need the finer grid. Each run takes about 10 s.
- **Composition:** set `concentration_mM` of the three Sample rows (potassium K+, sodium Na+, lithium Li+) to **0.1**.

Leave everything else at its default (Capillary length 30 mm, Applied voltage 500 V, Run time 20 s, Detector position 15 mm, EOF mobility 0). Press **Run CE simulation**.

Two facts about the injection that you will need: the sample plug is centered at **6 mm**, so the migration distance to a detector at 15 mm is ℓ = 9 mm. The plug has a Gaussian profile with a standard deviation of **σ_inj = 0.57 mm**.

Quantity	Where to find it	Value
Field E (V/m)	text line under the electropherogram	
Current density j (A/m²)	grey stamp under either plot (`j = …`)	
Joule heating (W/m)	text line under the electropherogram	
Background pH	text line under the electropherogram	

Analyte	Migration time t (s)	Peak width FWHM (s)	N = 5.54 (t / FWHM)²
potassium K+			
sodium Na+			
lithium Li+			

Q1. The power supply sets the voltage, and the buffer sets the current. Compute the conductivity of the background electrolyte from κ = j / E. Then check it: set **Overlay** to *Conductivity*, move the **Time slice** slider one step to the right of zero, and read the conductivity well away from the sample plug. (Skip slice 0, which shows the state before the buffer equilibrium is solved.) Do the two agree?

Part A: Where do the plates come from? (15 min)

If diffusion along the capillary were the only source of spreading, the plate count would be

> N = zeVℓ / (2 k_B T L), with k_B T / e = 0.02569 V at 298 K

where z is the charge, V the applied voltage, ℓ the migration distance and L the capillary length.

Q2. Predict first. Use the equation to predict N for K^+ at 500 V and at 1500 V. What does it say happens to N when the voltage triples?

Quantity	Your prediction
N for K^+ at 500 V (diffusion only)	
N for K^+ at 1500 V (diffusion only)	
Predicted ratio N(1500 V) / N(500 V)	

STOP: show your predictions to the instructor before running.

Q3. Run it. Set Applied voltage to 1000 V (Run time 12 s), run, then 1500 V (Run time 8 s), run.

Voltage (V)	t for K^+ (s)	FWHM for K^+ (s)	N for K^+	Joule heating (W/m)
500 (from the start table)				
1000				
1500				

Q4. Reconcile. Independent sources of spreading add as variances. For K^+ at 500 V, compute the diffusion variance $\sigma_{diff}^2 = 2Dt$, with $D = \mu_{ep} k_B T / e$ and $\mu_{ep} = 76.2 \times 10^{-9}$ $m^2 V^{-1} s^{-1}$. Compare it with σ_{inj}^2. Which one dominates? Use $N = \ell^2 / (\sigma_{inj}^2 + \sigma_{diff}^2)$ to predict N at 500 V and compare it with your measurement. Explain why the voltage barely mattered.

Q5. Predict. Return to 500 V. You will now move the **Detector position** from 15 mm to 25 mm (Run time 36 s). What is the new ℓ? Using the variance sum, predict N for K^+. Will the K^+ peak width in *seconds* grow, shrink or stay about the same? Why?

Quantity	Your prediction
New migration distance ℓ (mm)	
N for K^+	
FWHM for K^+ (↑ / ↓ / ≈), with reason	

STOP: show your predictions to the instructor before running.

Q6. Run and reconcile. Record t, FWHM and N for K^+ at the 25 mm detector. When the injection dominates the variance, what control actually raises N, and why does the width in seconds stay nearly fixed?

Detector (mm)	t for K^+ (s)	FWHM for K^+ (s)	N for K^+
25			

Part B: Peak shape from the sample's own conductivity (13 min)

Set Detector position back to 15 mm and Run time to 20 s. In the Composition table, change the **Tris** row into **imidazole**: name `imidazole`, mobility_e9 `52.0`, p_ka `7.05` (keep valence 1, 12 mM and role BGE). Run once with the samples still at 0.1 mM and record:

Analyte	μ_ep (×10^{-9})	t (s)	FWHM (s)	Peak shape (symmetric?)
potassium K+	76.2			
sodium Na+	51.9			
lithium Li+	40.1			

Imidazolium (μ = 52.0) is now the **co-ion**: the BGE cation that the sample cations displace to keep each zone electroneutral.

Q7. Predict. Each sample ion will be raised to **2 mM**. For each analyte, decide whether it is faster than, matched to, or slower than the co-ion. Then predict its peak shape and whether its time at the maximum moves earlier, later or stays the same.

Analyte	Faster / matched / slower than imidazolium	Predicted shape (diffuse front + sharp rear, symmetric, or sharp front + diffuse rear)	t at maximum (earlier / later / same)
K^+			
Na^+			
Li^+			

STOP: show your predictions to the instructor before running.

Q8. Run it. Set the three Sample concentrations to 2 and run.

Analyte	t (s)	FWHM (s)	Peak height (mM)	Observed shape
K^+				
Na^+				
Li^+				

Q9. Reconcile. Explain the K^+ peak in terms of conductivity and field: is the field inside the K^+ zone higher or lower than in the BGE? What happens to a K^+ ion that diffuses out ahead of the zone, and to one that falls behind? Then explain in one sentence why Na^+ stayed symmetric.

Part C: A salty sample (10 min)

Keep the imidazole BGE. Change the Composition so the sample is a trace of K^+ in a salt solution:

- potassium K+: `0.1` mM
- sodium Na+: `30` mM
- the lithium row becomes chloride: name `chloride Cl-`, valence `-1`, mobility_e9 `-79.1`, p_ka `-2`, `30` mM, role Sample

The app will warn that sodium and chloride exceed 10 % of the BGE. Run anyway. Chloride migrates toward the inlet and never reaches the detector, so ignore its row in the results table. The 30 mM sodium peak will dwarf the K^+ trace: click **sodium Na+** and **chloride Cl-** in the electropherogram legend to hide them, and the axis rescales to K^+.

Q10. Predict. Compared with K^+ at 0.1 mM in Part B (your first table), will the K^+ peak in the salty sample be wider, narrower or the same? Taller or shorter? Give the mechanism in one sentence.

STOP: show your predictions to the instructor before running.

Q11. Run it. Record the K^+ results. Then, with Overlay on *Conductivity* and Time slice one step right of zero, read the conductivity at the center of the sample plug and far from it.

Quantity	Value
K^+ t (s)	
K^+ FWHM (s)	
K^+ peak height (mM)	
κ in the sample plug (S/m)	
κ in the BGE (S/m)	

Q12. Reconcile. Compute FWHM(salty) / FWHM(Part B) and κ_sample / κ_BGE. Compare them using ℓ_zone ≈ ℓ_inj κ_sample / κ_BGE. What is this effect called? The app's warning names electromigration dispersion. Is that the same mechanism as Part B? Name one change to the sample preparation that would fix it.

Part D: pK_a, pH and migration order (6 min)

Keep the imidazole BGE (12 mM). Set Applied voltage to **1000 V** and Run time to **30 s**. In the Composition table, keep potassium K+ at 0.1 mM as a marker, and change the other two Sample rows:

name	valence	mobility_e9	p_ka	concentration_mM	role
pyridine	1	30.0	5.16	0.1	Sample
aniline	1	32.5	4.8	0.1	Sample

Pyridine and aniline are weak bases. Only the protonated form carries a charge, and the mobilities above are for the fully protonated cations. The fraction protonated and the effective mobility are

> α = 1 / (1 + 10^(pH − pK_a)), μ_eff = α μ_ep

Q13. Predict. When both are fully protonated, which cation is faster? Now use the background pH you recorded at the start (4.93) to fill in the table. The field at 1000 V is 33,333 V/m and ℓ = 9 mm.

Analyte	α at pH 4.93	μ_eff (×10⁻⁹)	Predicted t (s)
pyridine			
aniline			

Predicted order of arrival: _____

STOP: show your predictions to the instructor before running.

Q14. Run and reconcile. Record the migration times and the app's apparent mobilities. Did the order match your prediction? Explain in one sentence why a slower ion can arrive first.

Analyte	t (s)	Apparent mobility (×10⁻⁹)
pyridine		
aniline		

Q15. Extend. At some pH the two effective mobilities are equal and the peaks co-migrate. Find that pH by setting α μ_ep equal for the two bases. The BGE pH follows pH = 4.756 + log[c_im / (20 − c_im)], where c_im is the imidazole concentration in mM. What imidazole concentration gives the co-migration pH? Set it (round to the nearest mM), run, and describe what you see.

Class checkpoint (2 min)

Q16. Each Presenter gives one sentence: of the four effects you measured today (injection-limited plates, electromigration dispersion, antistacking, pH-controlled selectivity), which would most change how you prepare and run a real sample, and why?

Homework (individual)

The CE section has no CSV export, so **record the full settings for every run you report** (voltage, run time, detector position, grid points and the Composition table). Every number must be traceable to a stated set of conditions.

H1. Return to the Getting started conditions (Tris BGE, 0.1 mM samples, 500 V). Run the detector at 10, 15, 20 and 25 mm (Run time 20, 20, 30 and 36 s). Plot N for K^+ against ℓ^2. From the slope, estimate σ_inj and compare it with 0.57 mm.

H2. K^+ and NH_4^+ have nearly identical mobilities (76.2 and 76.1 $\times 10^{-9}$), but NH_4^+ is a weak acid with pK_a 9.25. Edit the five Composition rows to:

name	valence	mobility_e9	p_ka	concentration_mM	role
Tris	1	29.5	8.076	30	BGE
chloride Cl-	-1	-79.1	-2	10	BGE
potassium K+	1	76.2	14	0.1	Sample
ammonium NH4+	1	76.1	9.25	0.1	Sample
sodium Na+	1	51.9	14	0.1	Sample

Sodium stays in as a marker peak. Predict μ_eff for NH_4^+ at the pH the app reports, run at detector 15 mm (Run time 24 s) and 25 mm (Run time 44 s), and compute $R_s = 1.18\,\Delta t / (FWHM_1 + FWHM_2)$ for K^+ and NH_4^+. Repeat with chloride at 2 mM. Which pH separates the pair, and why? Look at the K^+ peak shape in the 2 mM chloride BGE and explain what changed.

H3. At 1500 V the app reported a Joule heating value. Compare it with the roughly 1 W/m that an air-cooled capillary can remove, and explain why no heating broadening appeared in Part A. Which two quantities would you change to make heating matter, and by what factor does the power per length grow for each?